REGISTER TRANSFER LANGUAGE

- Rather than specifying a digital system in words, a specific notation is used, *register transfer language*

- For any function of the computer, the register transfer language can be used to describe the (sequence of) microoperations

- Register transfer language
  - A symbolic language
  - A convenient tool for describing the internal organization of digital computers
  - Can also be used to facilitate the design process of digital systems.

DESIGNATION OF REGISTERS

- Registers are designated by capital letters, sometimes followed by numbers (e.g., A, R13, IR)
- Often the names indicate function:
  - MAR - memory address register
  - PC - program counter
  - IR - instruction register
- Registers and their contents can be viewed and represented in *various ways*
  - A register can be viewed as a single entity:

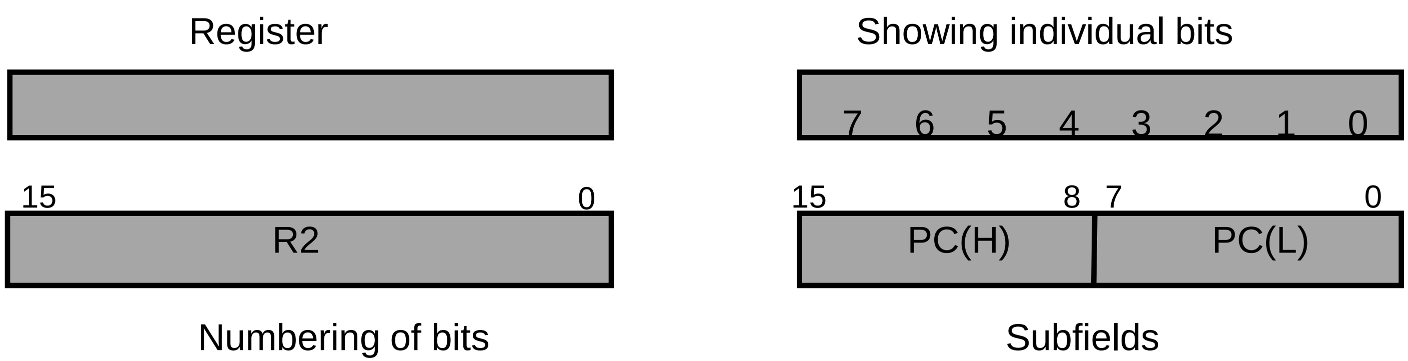
MAR

  - Registers may also be represented showing the bits of data they contain
  -

DESIGNATION OF REGISTERS

- Designation of a register
  - a register
  - portion of a register
  - a bit of a register

- Common ways of drawing the block diagram of a register



REGISTER TRANSFER

- Copying the contents of one register to another is a register transfer

- A register transfer is indicated as

$$R2 \leftarrow R1$$

  - In this case the contents of register R1 are copied (loaded) into register R2
  - A simultaneous transfer of all bits from the source R1 to the destination register R2, during one clock pulse
  - Note that this is a non-destructive; i.e. the contents of R1 are not altered by copying (loading) them to R2

REGISTER TRANSFER

- A register transfer such as

$$R3 \leftarrow R5$$

Implies that the digital system has

- the data lines from the source register (R5) to the destination register (R3)
- Parallel load in the destination register (R3)
- Control lines to perform the action

CONTROL FUNCTIONS

- Often actions need to only occur if a certain condition is true
- This is similar to an “if” statement in a programming language
- In digital systems, this is often done via a *control signal*, called a *control function*
  - If the signal is 1, the action takes place
- This is represented as:

$$P: R2 \leftarrow R1$$

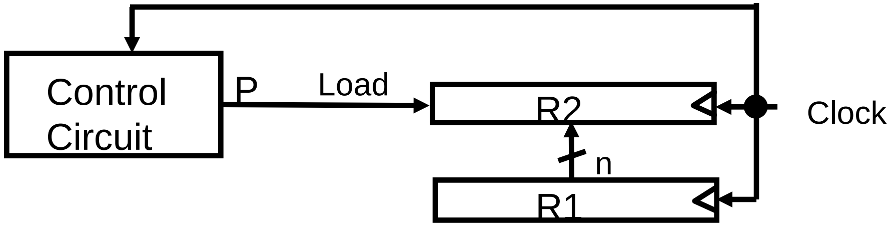
Which means “if P = 1, then load the contents of register R1 into register R2”, i.e., if (P = 1) then ($R2 \leftarrow R1$)

HARDWARE IMPLEMENTATION OF CONTROLLED TRANSFERS

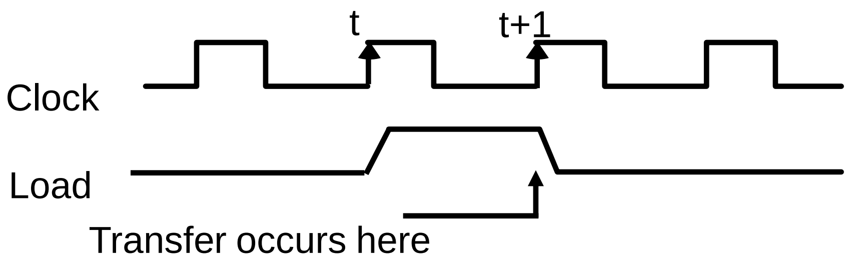
- Implementation of controlled transfer

P: R2 $\leftarrow$ R1

Block diagram



Timing diagram



- The same clock controls the circuits that generate the control function and the destination register
- Registers are assumed to use *positive-edge-triggered* flip-flops

SIMULTANEOUS OPERATIONS

- If two or more operations are to occur simultaneously, they are separated with commas
  - P: $R3 \leftarrow R5$, $MAR \leftarrow IR$
- Here, if the control function P = 1, load the contents of R5 into R3, and at the same time (clock), load the contents of register IR into register MAR